EECS 16A Designing Information Devices and Systems I
Homework 2

Submission Format
Your homework submission should consist of **one** file.

- `hw2.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned) as well as your IPython notebook saved as a PDF.

  If you do not attach a PDF "printout" of your IPython notebook, you will not receive credit for problems that involve coding. Make sure that your results and your plots are visible. Assign the IPython printout to the correct problem(s) on Gradescope.

Submit each file to its respective assignment on Gradescope.

1. Reading Assignment

For this homework, please read Notes 2B section 2.2, 3, 4, and 11A up to and including 11.3. These notes will give you an overview of linear transformations, linear independence, span, basic circuit elements, IV curves, and an introduction to thinking about and writing proofs. You are always welcome and encouraged to read beyond this as well. Write a paragraph about how you can use the strategies in the notes to tackle proof questions.

2. Easing into Proofs

(Contributors: Urmita Sikder, Gireeja Ranade)

Learning Objectives: *This is an opportunity to practice your proof development skills.*

(a) **Show that if the system of linear equations, $\mathbf{A}\vec{x} = \vec{b}$, has infinitely many solutions, then columns of A are linearly dependent.** Let us use the structure delineated in **Note 4** to approach this proof. This problem has 4 sub-parts and the following is a chart showing the sequential steps we are going to take to approach this proof.

In a text book you might see the steps in a proof written out in the order in the middle column of the table. But when you are building a proof you usually want to go in another order — this is the order of the subparts in this problem.

Proof steps		**Corresponding problem sub-parts**
1	Write what is known	Sub-part (i)
2	Manipulate what is known	Sub-part (iii)
3	Connecting it up	Sub-part (iv)
4	What is to be shown	Sub-part (ii)

(i) **First Step: write what you know**
Think about the *information we already know* from the problem statement. We know that system of equations, $\mathbf{A}\vec{x} = \vec{b}$, has infinitely many solutions. Infinitely many solutions are hard to work with, but perhaps we can simplify to something that we can work with. If the system has infinite number of solutions, it must have at least ___ distinct solutions (Fill in the blank).

So let us assume that $\vec{u}$ and $\vec{v}$ are two different vectors, both of which are solutions to $\mathbf{A}\vec{x} = \vec{b}$. Express the sentence above in a mathematical form (Just writing the equations will suffice; no need to do further mathematical manipulation).

(ii) **What we want to show:**
Now consider *what we need to show*. We have to show that the columns of $\mathbf{A}$ are linearly dependent. Let us assume that $\mathbf{A}$ has columns $\vec{c}_1$, $\vec{c}_2$, ..., and $\vec{c}_n$, i.e. $\mathbf{A} = \begin{bmatrix} | & | & \cdots & | \\ \vec{c}_1 & \vec{c}_2 & \cdots & \vec{c}_n \\ | & | & \cdots & | \end{bmatrix}$. Using the definition of linear dependence from **Note 3 Subsection 3.1.1**, write a mathematical equation that conveys linear dependence of $\vec{c}_1$, $\vec{c}_2$, ..., and $\vec{c}_n$.

(iii) **Manipulating what we know:**
Now let us try to start from the **First step: equations from (i)**, make mathematically logical steps and reach the **What we want to show: equations from (ii)**. Since your answer to (ii) is expressed in terms of the column vectors of $\mathbf{A}$, let us try to express the mathematical equations from (i), in terms of the the column vectors too. For example, we can write

$$\begin{aligned} & \mathbf{A}\vec{x} = \vec{b} \\ \implies & \begin{bmatrix} | & | & \cdots & | \\ \vec{c}_1 & \vec{c}_2 & \cdots & \vec{c}_n \\ | & | & \cdots & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdots \\ x_n \end{bmatrix} = \vec{b} \\ \implies & x_1\vec{c}_1 + x_2\vec{c}_2 + \ldots + x_n\vec{c}_n = \vec{b} \end{aligned}$$

Notice that x_1,... x_n etc are scalars. Now use your answer to part (i) to repeat the above formulation for distinct solutions $\vec{u}$ and $\vec{v}$. Note that this is proceeding slightly differently from how we did this proof in lecture. This is fine — there are often many correct ways to do a proof.

(iv) **Connecting it up:**
Now think about how you can mathematically manipulate your answer from part (iii) (**Manipulating what we know**) to **match the pattern** of your answer from part (ii) (**What we want to show**).

(b) **[PRACTICE]** Now try this proof on your own. Similar proofs will also be covered in your discussion section 3B. Given some set of vectors $\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$, show the following:

$$\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\} = \text{span}\{\vec{v}_1 + \vec{v}_2, \vec{v}_2, \ldots, \vec{v}_n\}$$

In order to show this, you have to prove the two following statements:

- If a vector $\vec{q}$ belongs in $\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$, then it must also belong in $\text{span}\{\vec{v}_1 + \vec{v}_2, \vec{v}_2, \ldots, \vec{v}_n\}$.
- If a vector $\vec{r}$ belong is $\text{span}\{\vec{v}_1 + \vec{v}_2, \vec{v}_2, \ldots, \vec{v}_n\}$, then it must also belong in $\text{span}\{\vec{v}_1, \vec{v}_2, \ldots, \vec{v}_n\}$.

In summary, you have to prove the problem statement from both directions. Now use the method developed in part (a) to prove these statements.

3. Image Stitching

(Contributors: Amanda Jackson, Ava Tan, Aviral Pandey, Gireeja Ranade, Lam Nguyen, Michael Kellman, Moses Won, Panagiotis Zarkos, Terry Chern, Titan Yuan, Urmita Sikder, Vijay Govindarajan)

Learning Objective: *This problem is similar to one that students might experience in an upper division computer vision course. Our goal is to give students a flavor of the power of tools from fundamental linear algebra and their wide range of applications.*

Often, when people take pictures of a large object, they are constrained by the field of vision of the camera. This means that they have two options to capture the entire object:

- Stand as far away as they need to include the entire object in the camera's field of view (clearly, we do not want to do this as it reduces the amount of detail in the image)
- (This is more exciting) Take several pictures of different parts of the object and stitch them together like a jigsaw puzzle.

We are going to explore the second option in this problem. Daniel, who is a professional photographer, wants to construct an image by using "image stitching". Unfortunately, Daniel took some of the pictures from different angles as well as from different positions and distances from the object. While processing these pictures, Daniel lost information about the positions and orientations from which the pictures were taken. Luckily, you and your friend Marcela, with your wealth of newly acquired knowledge about vectors and matrices, can help him!

You and Marcela are designing an iPhone app that stitches photographs together into one larger image. Marcela has already written an algorithm that finds common points in overlapping images. **It's your job to figure out how to stitch the images together using Marcela's common points to reconstruct the larger image.**

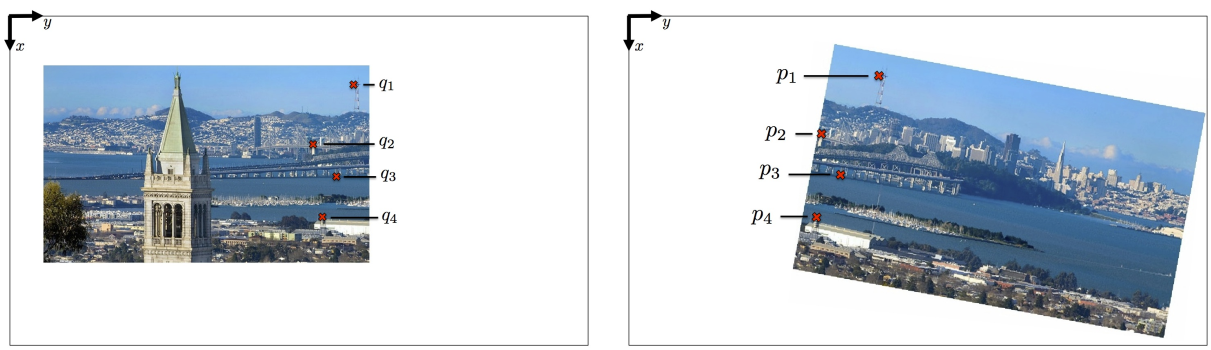

Figure 1: Two images to be stitched together with pairs of matching points labeled.

We will use vectors to represent the common points which are related by a linear transformation. Your idea is to find this linear transformation. For this you will use a single matrix, $\mathbf{R}$, and a vector, $\vec{t}$, that transforms every common point in one image to their corresponding point in the other image. Once you find $\mathbf{R}$ and $\vec{t}$ you will be able to transform one image so that it lines up with the other image.

Suppose $\vec{p} = \begin{bmatrix} p_x \\ p_y \end{bmatrix}$ is a point in one image, which is transformed to $\vec{q} = \begin{bmatrix} q_x \\ q_y \end{bmatrix}$, the corresponding point in the other image (i.e., they represent the same object in the scene). For example, Fig. 1 shows how the points $\vec{p}_1$, $\vec{p}_2$ ... in the right image are transformed to points $\vec{q}_1$, $\vec{q}_2$ ... on the left image. You write down the following relationship between $\vec{p}$ and $\vec{q}$.

$$\begin{bmatrix} q_x \\ q_y \end{bmatrix} = \underbrace{\begin{bmatrix} r_{xx} & r_{xy} \\ r_{yx} & r_{yy} \end{bmatrix}}_{\mathbf{R}} \begin{bmatrix} p_x \\ p_y \end{bmatrix} + \underbrace{\begin{bmatrix} t_x \\ t_y \end{bmatrix}}_{\vec{t}} \tag{1}$$

This problem focuses on finding the unknowns (i.e. the components of $\mathbf{R}$ and $\vec{t}$), so that you will be able to stitch the image together.

(a) To understand how the matrix $\mathbf{R}$ and vector $\vec{t}$ transforms any vector representing a point on a image, Consider this equation similar to Equation (1),

$$\vec{v} = \begin{bmatrix} 2 & 2 \\ -2 & 2 \end{bmatrix} \vec{u} + \vec{w} = \vec{v}_1 + \vec{w}. \tag{2}$$

Use $\vec{w} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ for this part.

We want to find out what geometric transformation(s) can be applied on $\vec{u}$ to give $\vec{v}$.

Step 1: Find out how $\begin{bmatrix} 2 & 2 \\ -2 & 2 \end{bmatrix}$ is transforming $\vec{u}$. Evaluate $\vec{v}_1 = \begin{bmatrix} 2 & 2 \\ -2 & 2 \end{bmatrix} \vec{u}$.

What **geometric transformation(s)** might be applied to $\vec{u}$ to get $\vec{v}_1$? Choose the option that answers the question and explain your choice.

(i) Rotation
(ii) Scaling
(iii) Shifting/Translation
(iv) Rotation and Scaling

Drawing the vectors $\vec{u}$, and $\vec{v}_1$ in two dimensions on a single plot might help you to visualize the transformations. You can also look into the corresponding demo in the IPython notebook `prob2.ipynb`.

Step 2: Find out $\vec{v} = \vec{v}_1 + \vec{w}$. Find out how **addition of $\vec{w}$ is geometrically transforming** $\vec{v}_1$. Choose the option that answers the question and explain your choice.

(i) Rotation
(ii) Scaling
(iii) Shifting/ Translation.

Drawing the vectors $\vec{v}$, $\vec{w}$, and $\vec{v}_1$ in two dimensions on a single plot might help you to visualize the transformations. You can also look into the corresponding demo in the IPython notebook `prob2.ipynb`.

(b) Multiply Equation (1) out into **two scalar linear equations**.

(i) What are the known values and what are the unknowns in each equation?
(ii) How many unknowns are there?
(iii) How many independent equations do you need to solve for all the unknowns?
(iv) How many pairs of common points $\vec{p}$ and $\vec{q}$ will you need in order to write down a system of equations that you can use to solve for the unknowns? ***Hint:*** *Remember that each pair of $\vec{p}$ and $\vec{q}$ will give you* ***two*** *different linear equations.*

(c) Write out a system of linear equations that you can use to solve for $\vec{\alpha} = \begin{bmatrix} r_{xx} \\ r_{xy} \\ r_{yx} \\ r_{yy} \\ t_x \\ t_y \end{bmatrix}$. Assume that all four pairs of points from Fig. 1 are labeled as:

$$\vec{q}_1 = \begin{bmatrix} q_{1x} \\ q_{1y} \end{bmatrix}, \vec{p}_1 = \begin{bmatrix} p_{1x} \\ p_{1y} \end{bmatrix} \qquad \vec{q}_2 = \begin{bmatrix} q_{2x} \\ q_{2y} \end{bmatrix}, \vec{p}_2 = \begin{bmatrix} p_{2x} \\ p_{2y} \end{bmatrix} \qquad \vec{q}_3 = \begin{bmatrix} q_{3x} \\ q_{3y} \end{bmatrix}, \vec{p}_3 = \begin{bmatrix} p_{3x} \\ p_{3y} \end{bmatrix} \qquad \vec{q}_4 = \begin{bmatrix} q_{4x} \\ q_{4y} \end{bmatrix}, \vec{p}_4 = \begin{bmatrix} p_{4x} \\ p_{4y} \end{bmatrix}.$$

Now think of your answer to Part b(iv). How many pairs of these points would you need to solve for $\vec{\alpha}$. Choose **just enough** equations required to solve for $\vec{\alpha}$ and express these linear equations using **matrix-vector form**.

(d) In the IPython notebook `prob2.ipynb`, you will have a chance to test out your solution. Plug in the values that you are given for p_x, p_y, q_x, and q_y for each pair of points into your system of equations to solve for the matrix, $\mathbf{R}$, and vector, $\vec{t}$. The notebook will solve the system of equations, apply your transformation to the second image, and show you if your stitching algorithm works. **You are NOT responsible for understanding the image stitching code or Marcela's algorithm.**

4. Kinematic Model for a Simple Car

Learning Objective: *Many real world systems are not actually linear and have more complex behaviors. However, linear models can approximate non-linear systems under certain conditions.*

Building a self-driving car first requires understanding the basic motions of a car. In this problem, we will explore how to model the motion of a car.

There are several models that we can use to model the motion of a car. Assume we use a kinematic model, described in the following four equations and Figure 2.

$$x[k+1] = x[k] + v[k]\cos(\theta[k])\Delta t \tag{3}$$
$$y[k+1] = y[k] + v[k]\sin(\theta[k])\Delta t \tag{4}$$
$$\theta[k+1] = \theta[k] + \frac{v[k]}{L}\tan(\phi[k])\Delta t \tag{5}$$
$$v[k+1] = v[k] + a[k]\Delta t \tag{6}$$

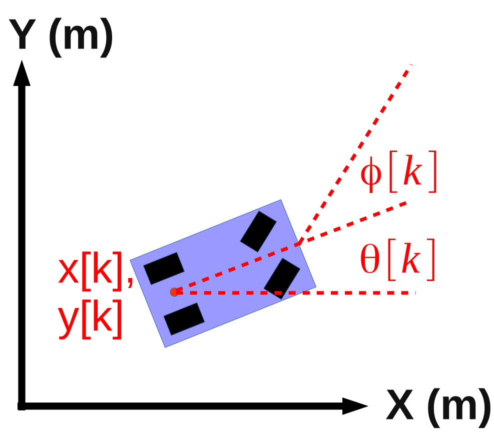

Figure 2: Vehicle Kinematic Model

where

- k, a nonnegative integer, indicates the time step at which we measure each variable (e.g. $v[k]$ is the speed at time step k and $v[k+1]$ is the speed at the following time step)
- $x[k]$ and $y[k]$ denote the coordinates of the vehicle (meters)
- $\theta[k]$ denotes the heading of the vehicle, or the angle with respect to the x-axis (radians)

- $v[k]$ is the speed of the car (meters per second)
- $a[k]$ is the acceleration of the car (meters per second squared)
- $\phi[k]$ is the steering angle input we command (radians)
- Δt is a constant measuring the time difference (in seconds) between time steps $k+1$ and k
- L is a constant and is the length of the car (in meters)

For this problem, let L be 1.0 meter and Δt be 0.1 seconds.

The variables $x[k], y[k], \theta[k], v[k]$ describe the **state** of the car at time step k. The state captures all the information needed to fully determine the current position, speed, and heading of the car. The **inputs** at time step k are $a[k]$ and $\phi[k]$. These are provided by the driver. The current value of these inputs, along with the current state of the vehicle, will determine the state of the vehicle at the next time step.

We note that the problem is nonlinear, due to the sine, cosine and tangent functions, as well as terms including the product of states and inputs.

The purpose of this problem is to show that we can approximate a nonlinear model with a simple linear model and do reasonably well. This is why, despite many systems being nonlinear, linear algebra tools are widely used in practice.

For Parts (b) - (d), fill out the corresponding sections in prob2.ipynb.

(a) We assume that the car has a small heading, θ, which is a **very small but nonzero** value, and that the steering angle ϕ is also **very small but nonzero**. In this case, we could use the following approximations:

$$\begin{aligned} \sin(\alpha) &\approx 0, \\ \cos(\alpha) &\approx 1, \\ \tan(\alpha) &\approx 0. \end{aligned}$$

where α is the small angle of interest, and $\approx$ means "approximately equal to". Here, we use a very simple approximation for small angles; in later classes, you may learn better approximations.

Draw, by hand, graphs of $\sin(\alpha)$ and $\cos(\alpha)$, for α ranging from $-\pi$ to π. Using these graphs, can you justify the approximation we are making for small values of α?

(b) Applying the approximation described in the previous part, write down a system of linear equations that approximates the nonlinear vehicle model given above in Equations (3) to (6). In particular, find the 4 x 4 matrix **A** and 4 x 2 matrix **B** that satisfy the equation given below.

$$\begin{bmatrix} x[k+1] \\ y[k+1] \\ \theta[k+1] \\ v[k+1] \end{bmatrix} = \mathbf{A} \begin{bmatrix} x[k] \\ y[k] \\ \theta[k] \\ v[k] \end{bmatrix} + \mathbf{B} \begin{bmatrix} a[k] \\ \phi[k] \end{bmatrix}$$

Hint: *Start with simplifying Equations* (3) *to* (6)*, using the* $\sin(\alpha)$*,* $\cos(\alpha)$*, and* $\tan(\alpha)$ *approximations from part (a) only to approximate nonlinear terms. Do NOT replace* ϕ*,* θ *with 0 unless it is for the* $\sin(\alpha)$*,* $\cos(\alpha)$*, and* $\tan(\alpha)$ *approximations.*

(c) Suppose we drive the car so that the direction of travel is aligned with the x-axis, and we are driving nearly straight, i.e. the steering angle is $\phi[k] = 0.0001$ radians. (Driving exactly straight would have the steering angle $\phi[k] = 0$ radians.) The initial state and input are:

$$\begin{bmatrix} x[0] \\ y[0] \\ \theta[0] \\ v[0] \end{bmatrix} = \begin{bmatrix} 5.0 \\ 10.0 \\ 0.0 \\ 2.0 \end{bmatrix}$$

$$\begin{bmatrix} a[k] \\ \phi[k] \end{bmatrix} = \begin{bmatrix} 1.0 \\ 0.0001 \end{bmatrix}$$

You can use these values in the IPython notebook to compare how the nonlinear system evolves in comparison to the linear approximation that you made. The IPython notebook simulates the car for ten time steps. Are the trajectories similar or very different? Why?

(d) Now suppose we drive the vehicle from the same starting state, but we turn left instead of going straight, i.e. the steering angle is $\phi[k] = 0.5$ radians. The initial state and input are:

$$\begin{bmatrix} x[0] \\ y[0] \\ \theta[0] \\ v[0] \end{bmatrix} = \begin{bmatrix} 5.0 \\ 10.0 \\ 0.0 \\ 2.0 \end{bmatrix}$$

$$\begin{bmatrix} a[k] \\ \phi[k] \end{bmatrix} = \begin{bmatrix} 1.0 \\ 0.5 \end{bmatrix}$$

You can use these values in the IPython notebook to compare how the nonlinear system evolves in comparison to the linear approximation that you made. The IPython notebook simulates the car for ten time steps. Are the trajectories similar or very different? Why?

5. Basic Circuit Components

Learning Objectives: *Review basics of cucuit components and current-voltage relationships*

In the laboratory, you are tasked with identifying a single unknown component within a circuit. You can use a piece of electrical equipment called a multimeter to measure either voltage (voltmeter) across or the current (ammeter) through the component (you can measure both quantities simultaneously using two multimeters).

For each part of the problem, deduce the most likely type of circuit component based on the provided voltage and current measurements. Also draw the component circuit symbol and sketch the IV curve.

Hint: You are told the possible choices are ***short circuit (wire)***, ***open circuit***, ***resistor***, ***voltage source***, *and* ***current source***. *Moreover, each part in this problem has a unique component, there are no repeats.*

(a) First, to familiarize yourself with common quantities of *voltage*, *current*, and *resistance*, fill in the unit name and unit symbol for each:

Quantity	**Symbol**	**Unit Name**	**Unit Symbol**
Voltage	V		
Current	I		
Resistance	R		

(b) You take one measurement and find the voltage is $V = 10$ V and current is $I = 1$ A. After changing a part of the circuit (not the part you are measuring), you take another measurement and find $V = 10$ V and $I = 2$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

(c) You take one measurement and find the voltage is $V = 0$ V and current is $I = 1$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

(d) You take one measurement and find the voltage is $V = 5$ V and current is $I = 0$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

(e) You take one measurement and find the voltage is $V = 10$ V and current is $I = 2$ A. After changing a part of the circuit (not the part you are measuring), you take another measurement and find $V = -5$ V and $I = -1$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

(f) You take one measurement and find the voltage is $V = 10$ V and current is $I = 2$ A. After changing a part of the circuit (not the part you are measuring), you take another measurement and find $V = -3$ V and $I = 2$ A. What is the most likely component type? Draw the component symbol and sketch the IV curve.

6. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.